

GPIE

Green Policy Intelligence Engine

All Maps and Plots

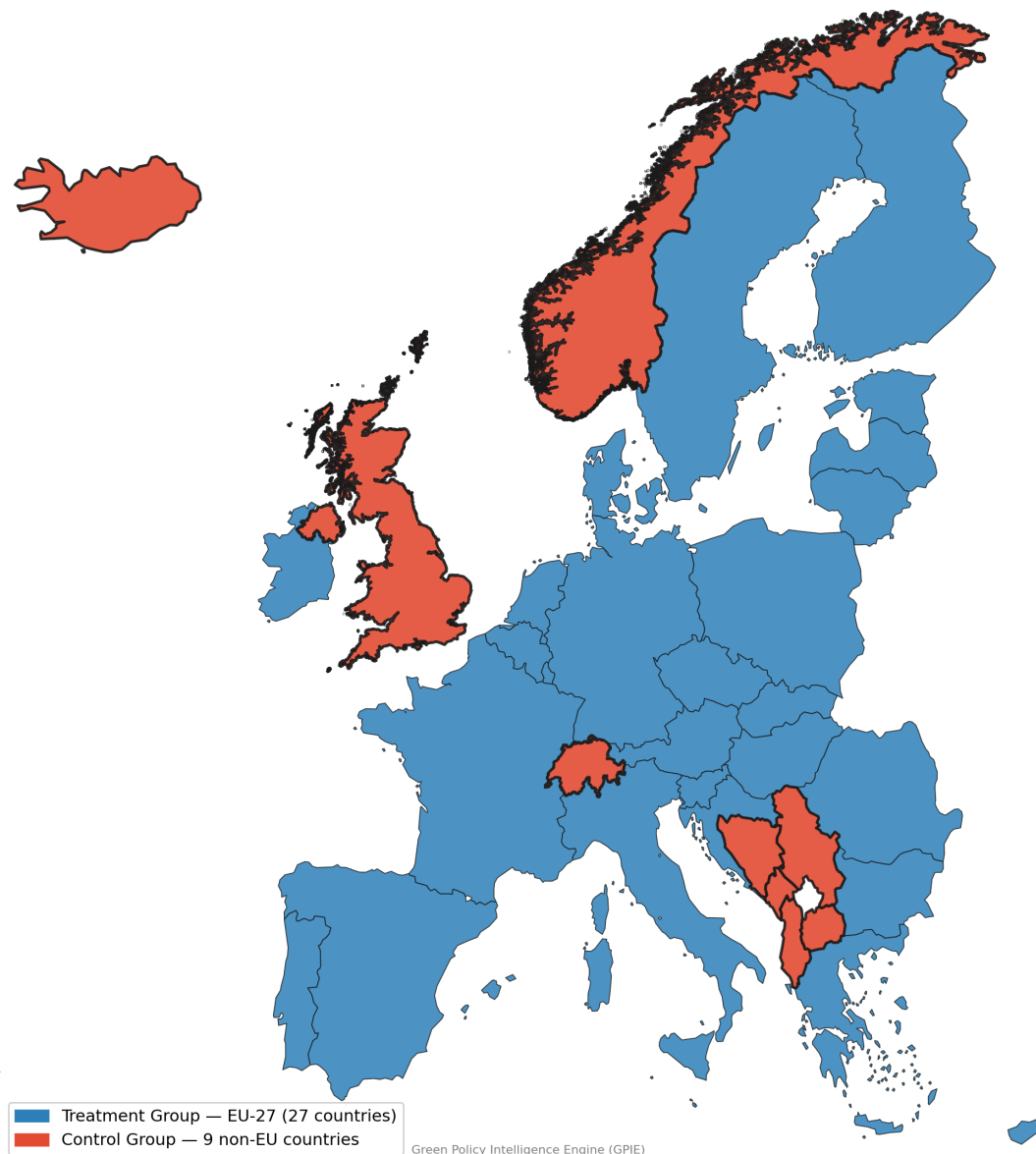
18 figures compiled from outputs/plots/

1. Study Design: Treatment vs. Control Group
2. NO₂ Choropleth Map
3. NO₂ Before vs. After the European Climate Law
4. NO₂: EU-27 vs. Control Group
5. Event-Study: Quarter-by-Quarter NO₂ Effect
6. NDVI Choropleth Map
7. NDVI Before vs. After the European Climate Law
8. NDVI: EU-27 vs. Control Group
9. GDP Choropleth Map
10. Dominant Land Cover Class Map
11. Elevation (DEM) Map
12. Climate / Temperature Map
13. India Transferability Validation
14. Augmented Synthetic Control: EU-27 vs. 7-Country Donor Composite
15. Local Moran's I (LISA) Spatial Cluster Map
16. Policies by Year
17. Policy Type Distribution
18. Policy Types by Year

Study Design: Treatment vs. Control Group

EU-27 (treatment) and the 9-country non-EU control group - UK, Norway, Switzerland, Iceland, Albania, Bosnia and Herzegovina, Montenegro, North Macedonia, Serbia.

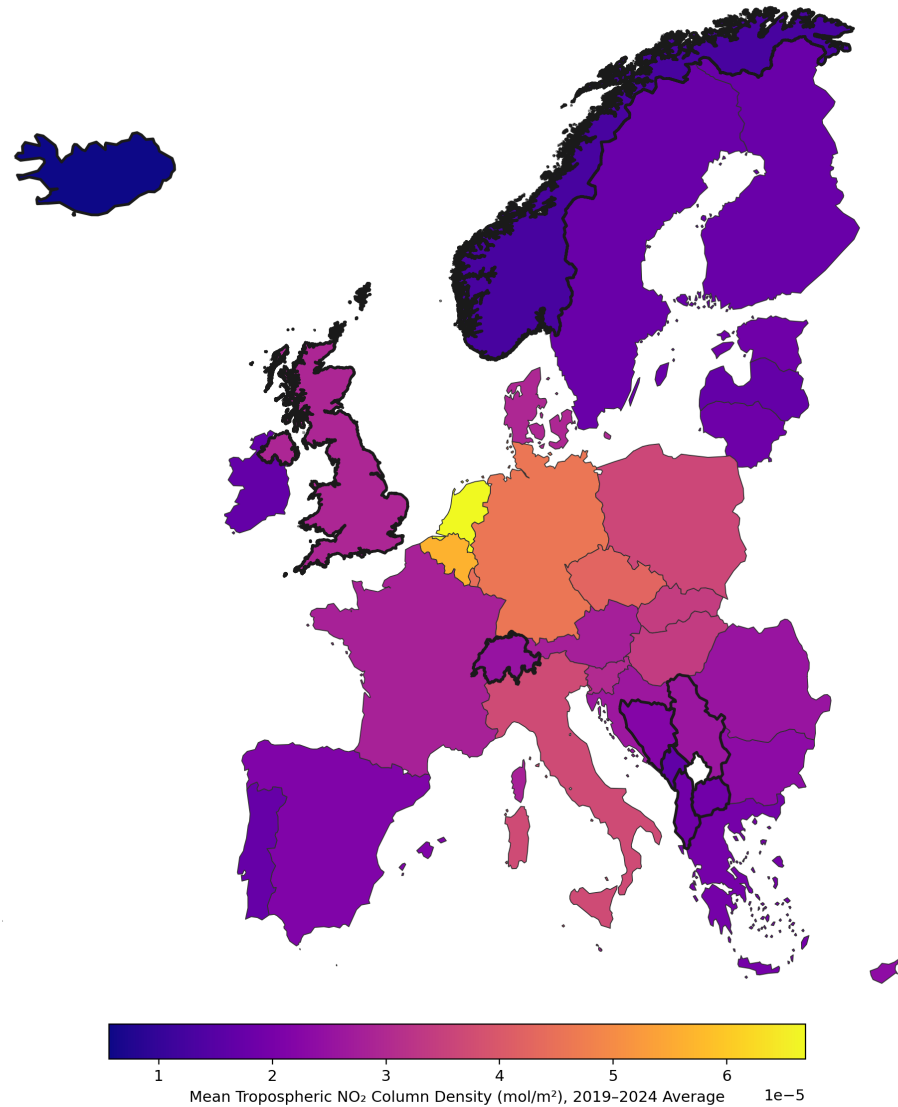
GPIE Study Design: Treatment vs. Control Group
Difference-in-Differences comparison — EU-27 (subject to the European Climate Law) vs. nine non-EU comparator countries
UK, Norway, Switzerland, Iceland, Albania, Bosnia and Herzegovina, Montenegro, North Macedonia, Serbia



NO2 Choropleth Map

Mean tropospheric NO₂ concentration by country, Sentinel-5P TROPOMI.

Tropospheric NO₂ Concentration: EU-27 vs. 9-Country Control Group
Thick borders mark non-EU control-group countries — visually comparable NO₂ levels support the DiD null result



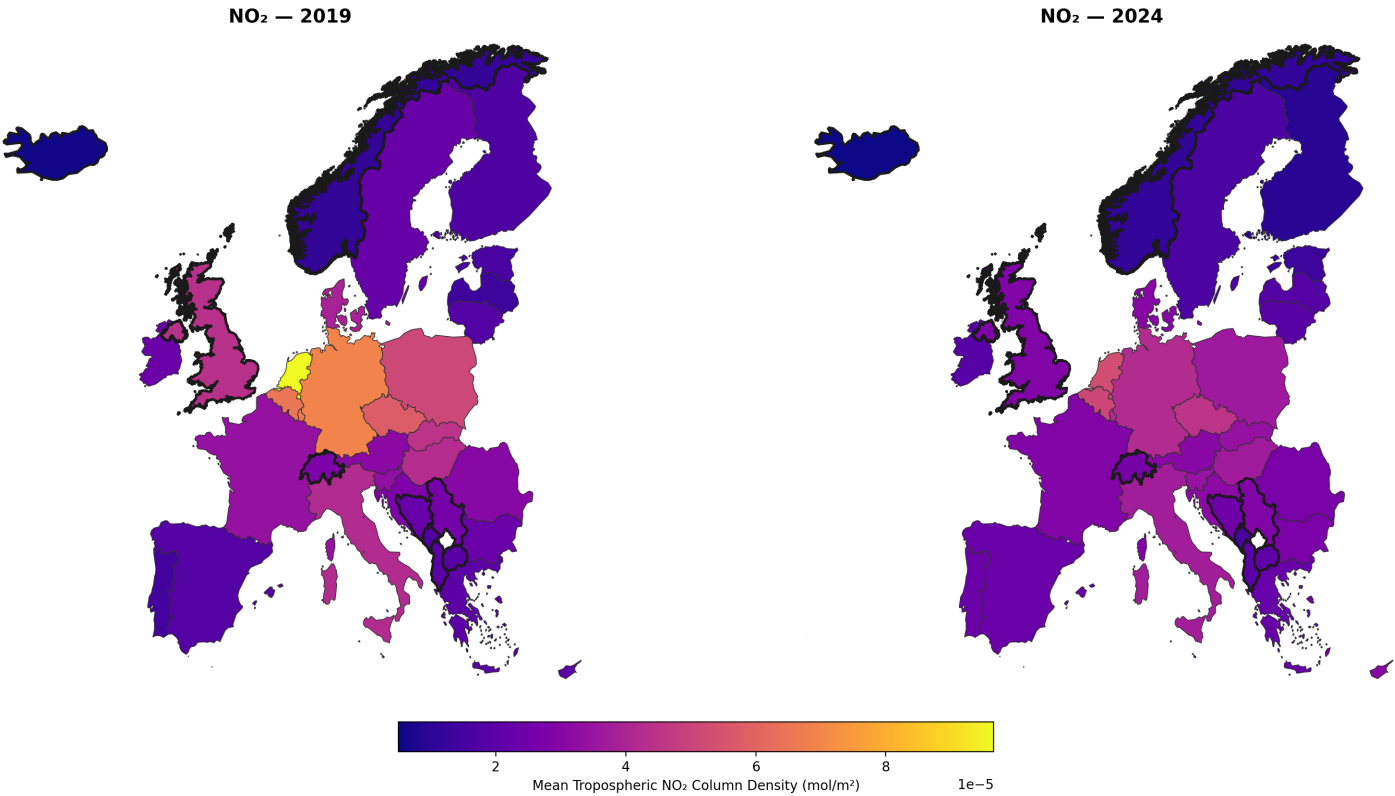
Green Policy Intelligence Engine (GPiE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

GPiE - Green Policy Intelligence Engine

NO2 Before vs. After the European Climate Law

Average NO2 across the EU-27, 2019 (pre-treatment) vs. 2024 (post-treatment).

NO₂ Levels Before vs. After the European Climate Law: EU-27 and Control Group (2019 vs. 2024)
Thick borders mark the 9-country non-EU control group — comparable decline pattern across both groups

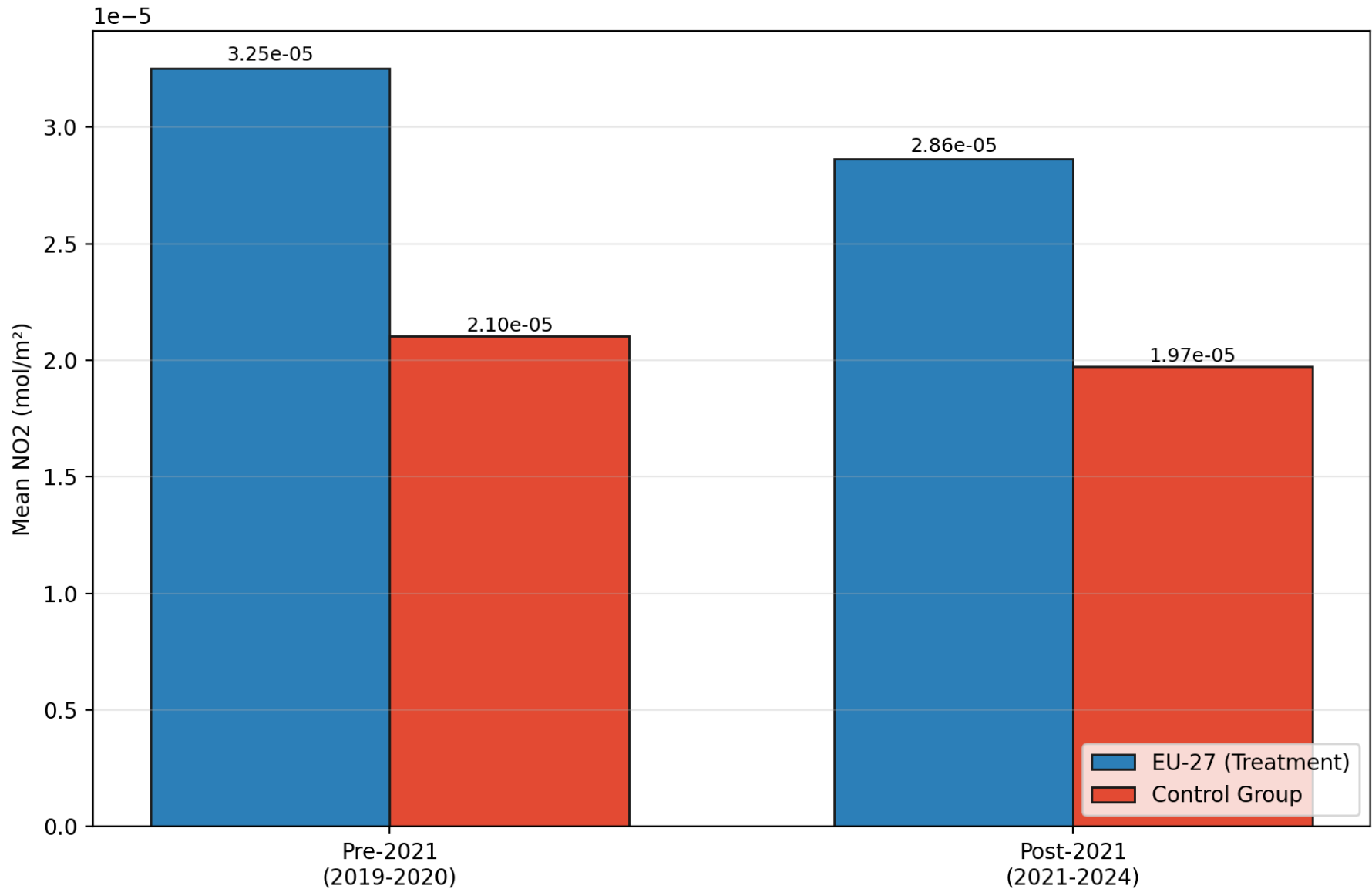


Green Policy Intelligence Engine (GPiE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

NO2: EU-27 vs. Control Group

Mean NO2, pre- and post-treatment, EU-27 vs. 9-country control group - coefficient = $-2.22\text{e-}06$, $p = 0.101$.

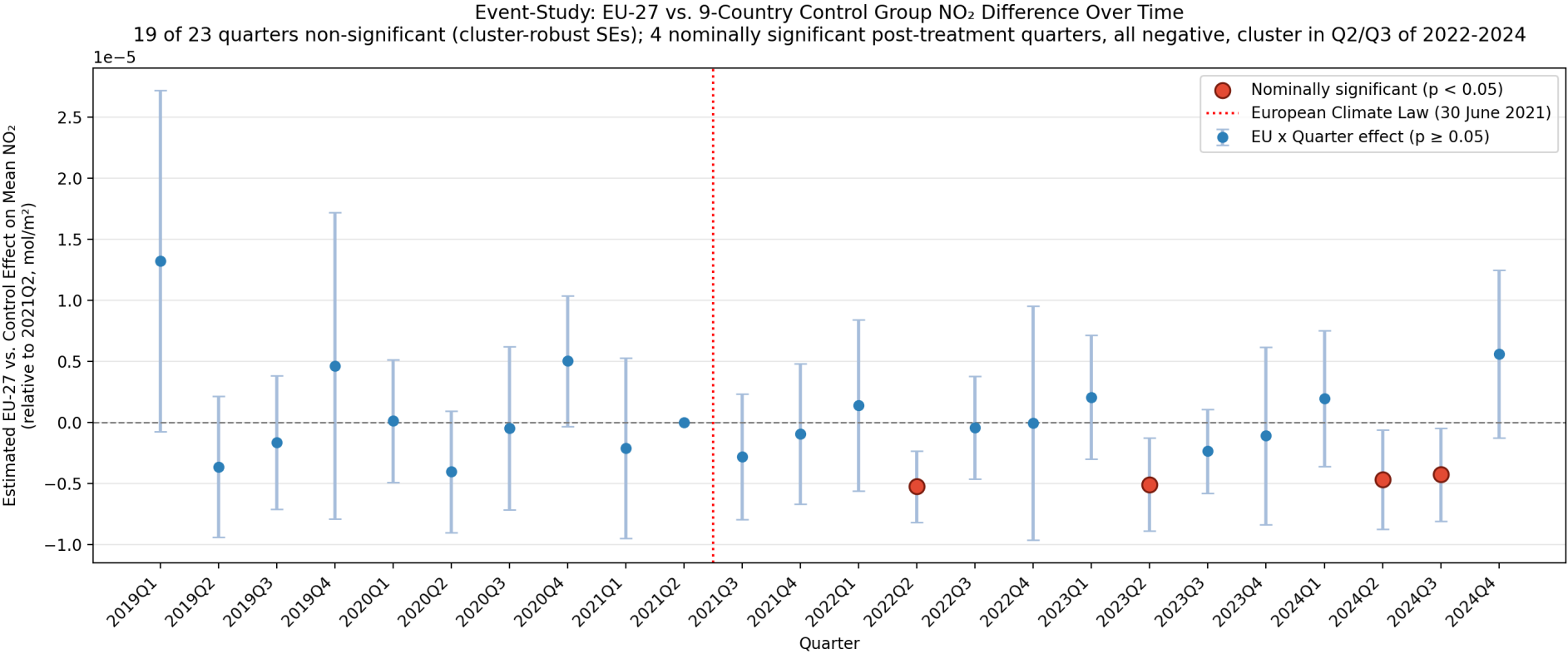
NO2: EU-27 vs. 9-Country Control Group, Before vs. After the European Climate Law
Two-group DiD model: coefficient = $-2.22\text{e-}06$, $p = 0.101$ (cluster-robust) -
no significant pooled effect, though a baseline-pollution split finds one in higher-baseline countries



Green Policy Intelligence Engine (GPIE) - Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

Event-Study: Quarter-by-Quarter NO2 Effect

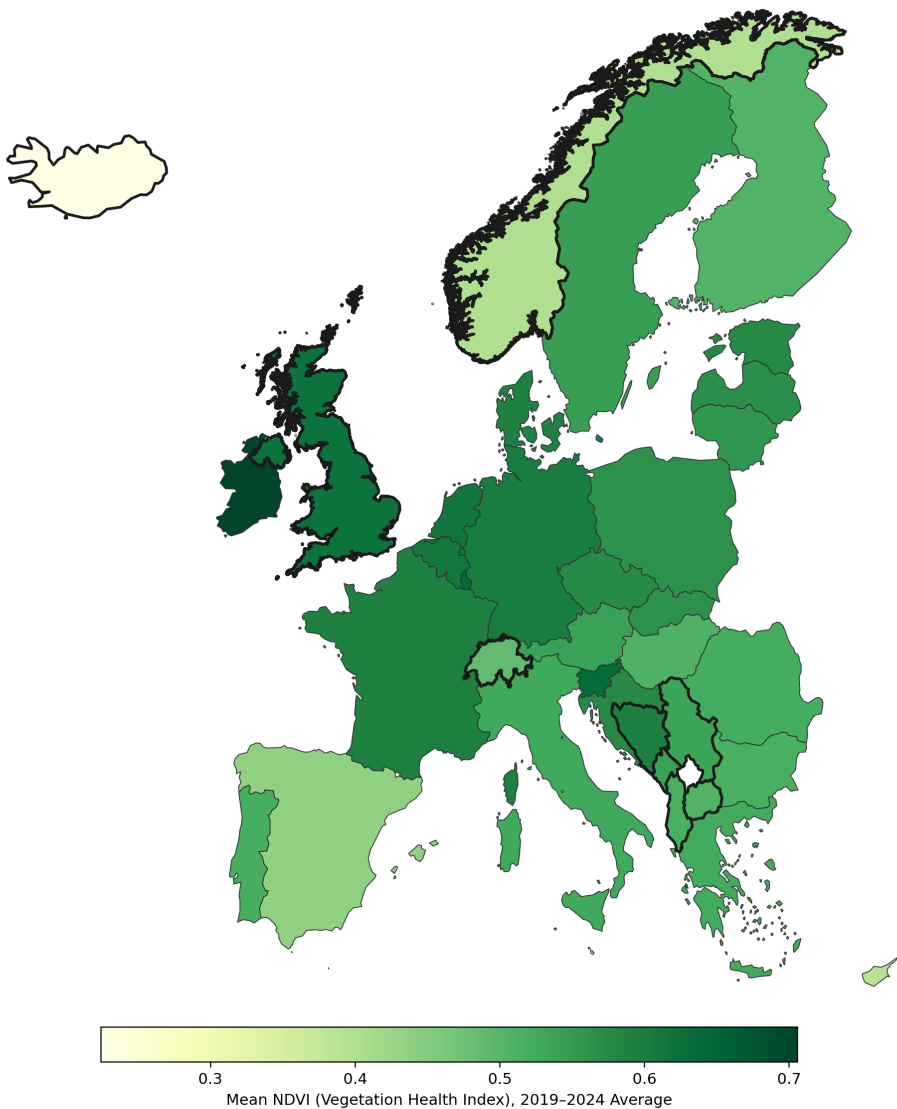
23-quarter Difference-in-Differences event-study estimates, cluster-robust SEs - 4 significant post-treatment quarters, all negative, clustering in Q2/Q3 of 2022-2024.



NDVI Choropleth Map

Mean vegetation health index (NDVI) by country.

Vegetation Health (NDVI): EU-27 vs. 9-Country Control Group
Thick borders mark non-EU control-group countries



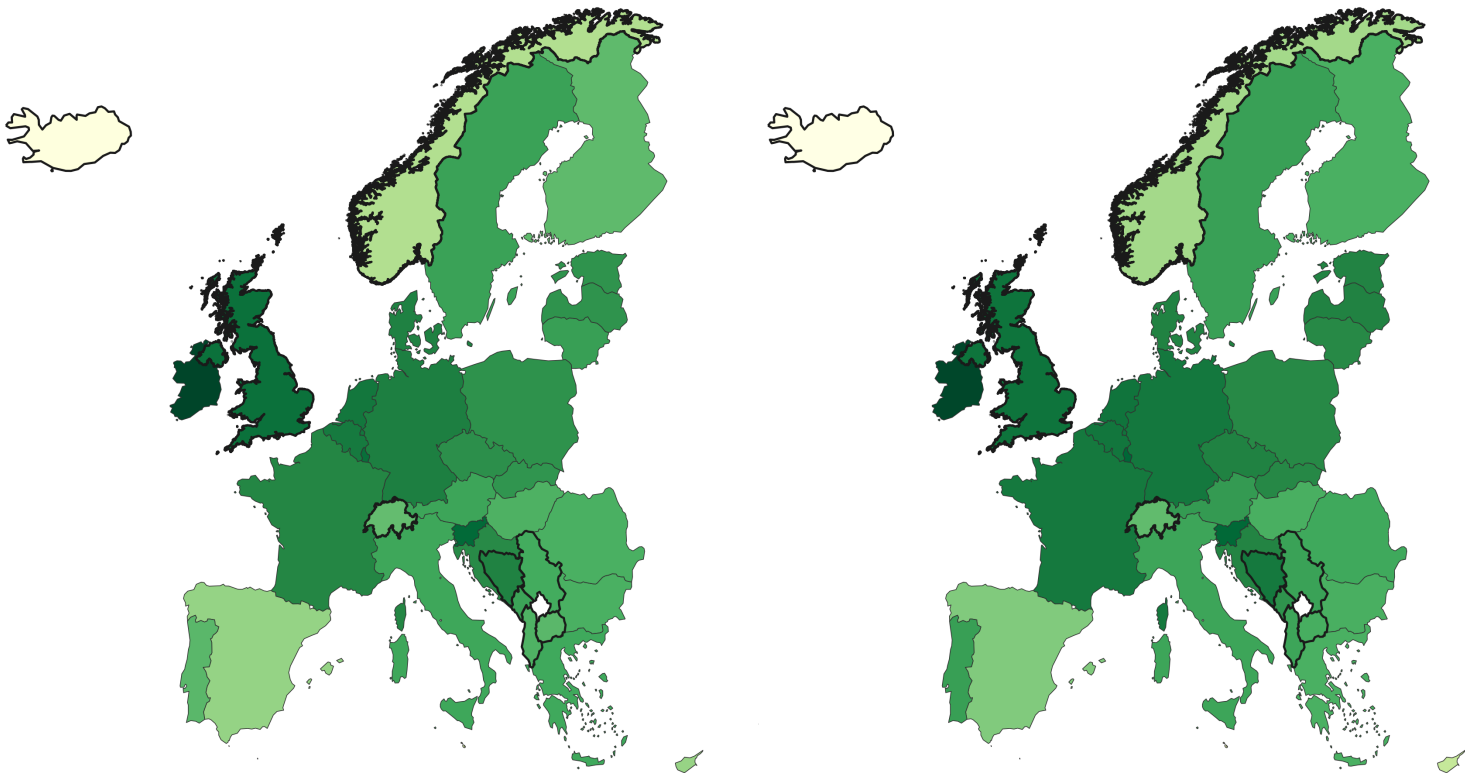
NDVI Before vs. After the European Climate Law

Average NDVI across the EU-27, pre- vs. post-treatment.

NDVI (Vegetation Health) Before vs. After the European Climate Law: EU-27 and Control Group (2019 vs. 2024)
Thick borders mark the 9-country non-EU control group

NDVI — 2019

NDVI — 2024

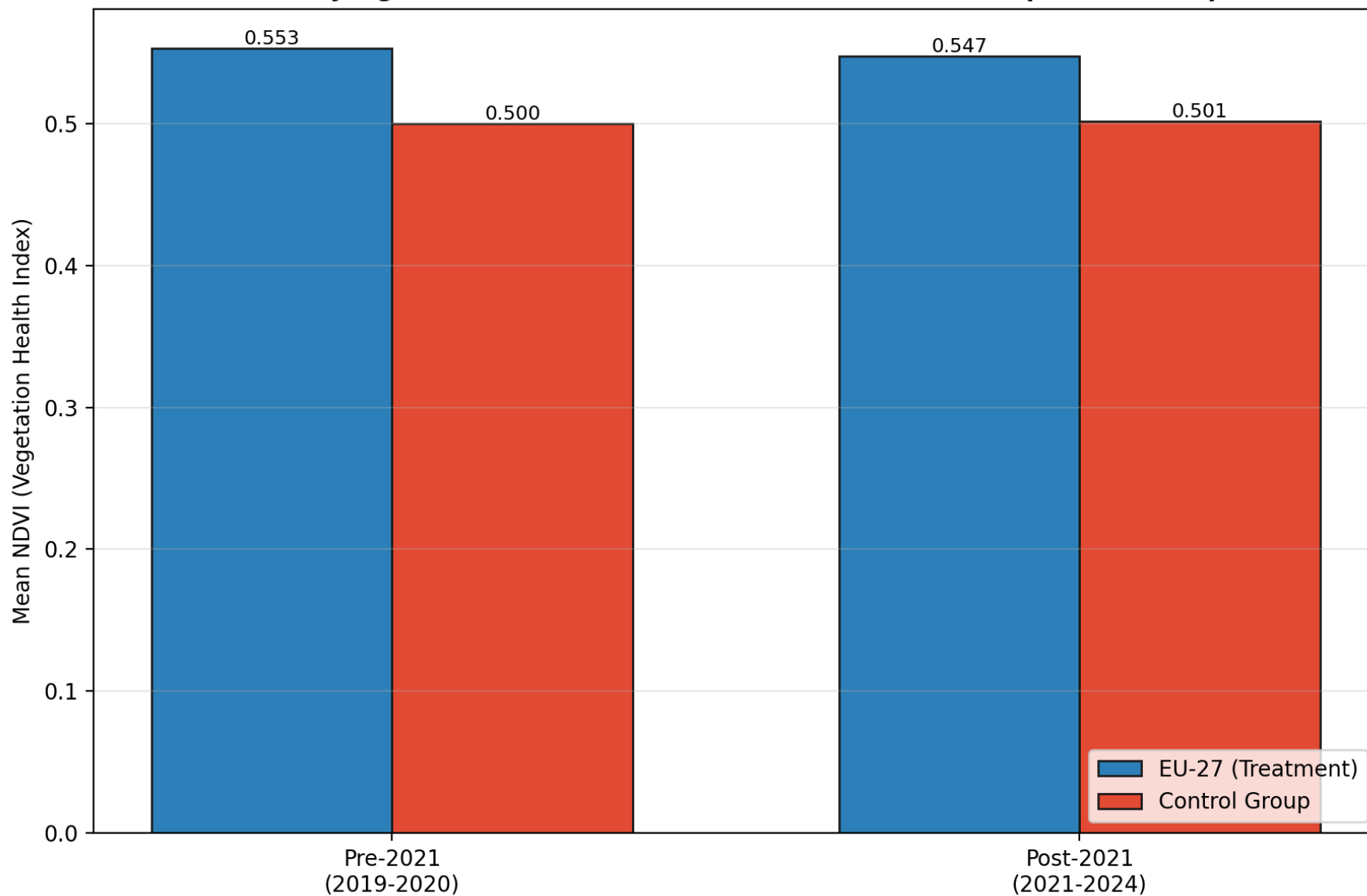


Green Policy Intelligence Engine (GPIE) — Source: CGLS NDVI 300m, Sentinel Hub Statistical API

NDVI: EU-27 vs. Control Group

Two-group DiD model: coefficient = -0.0145, p = 0.007 (cluster-robust) - a statistically significant relative decline.

NDVI: EU-27 vs. 9-Country Control Group, Before vs. After the European Climate Law
Two-group DiD model: coefficient = -0.0145, p = 0.007 (cluster-robust) -
a statistically significant relative decline, not visible in NO2's equivalent comparison



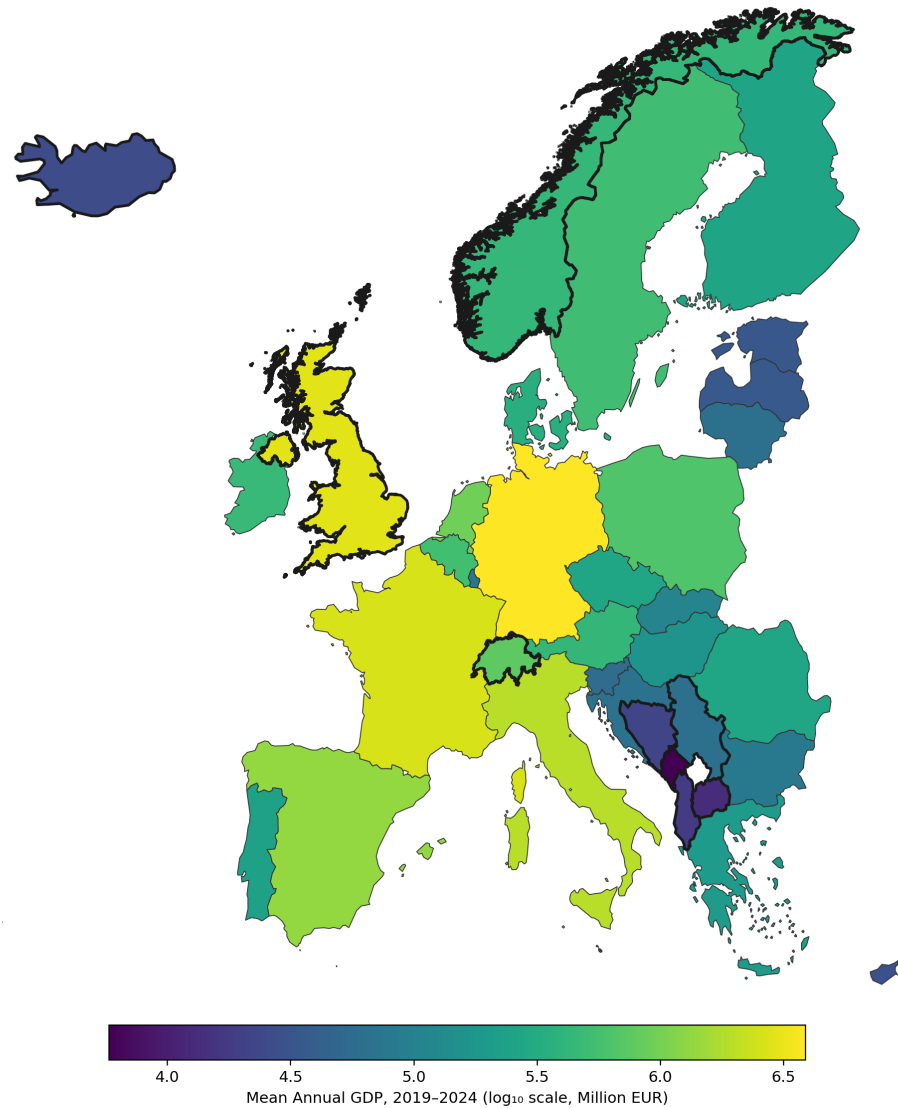
Green Policy Intelligence Engine (GPIE) - Source: CGLS NDVI 300m, Sentinel Hub Statistical API

GDP Choropleth Map

Control variable - GDP by country.

GDP: EU-27 vs. 9-Country Control Group

Thick borders mark non-EU control-group countries — GDP used as an economic control variable in the causal model



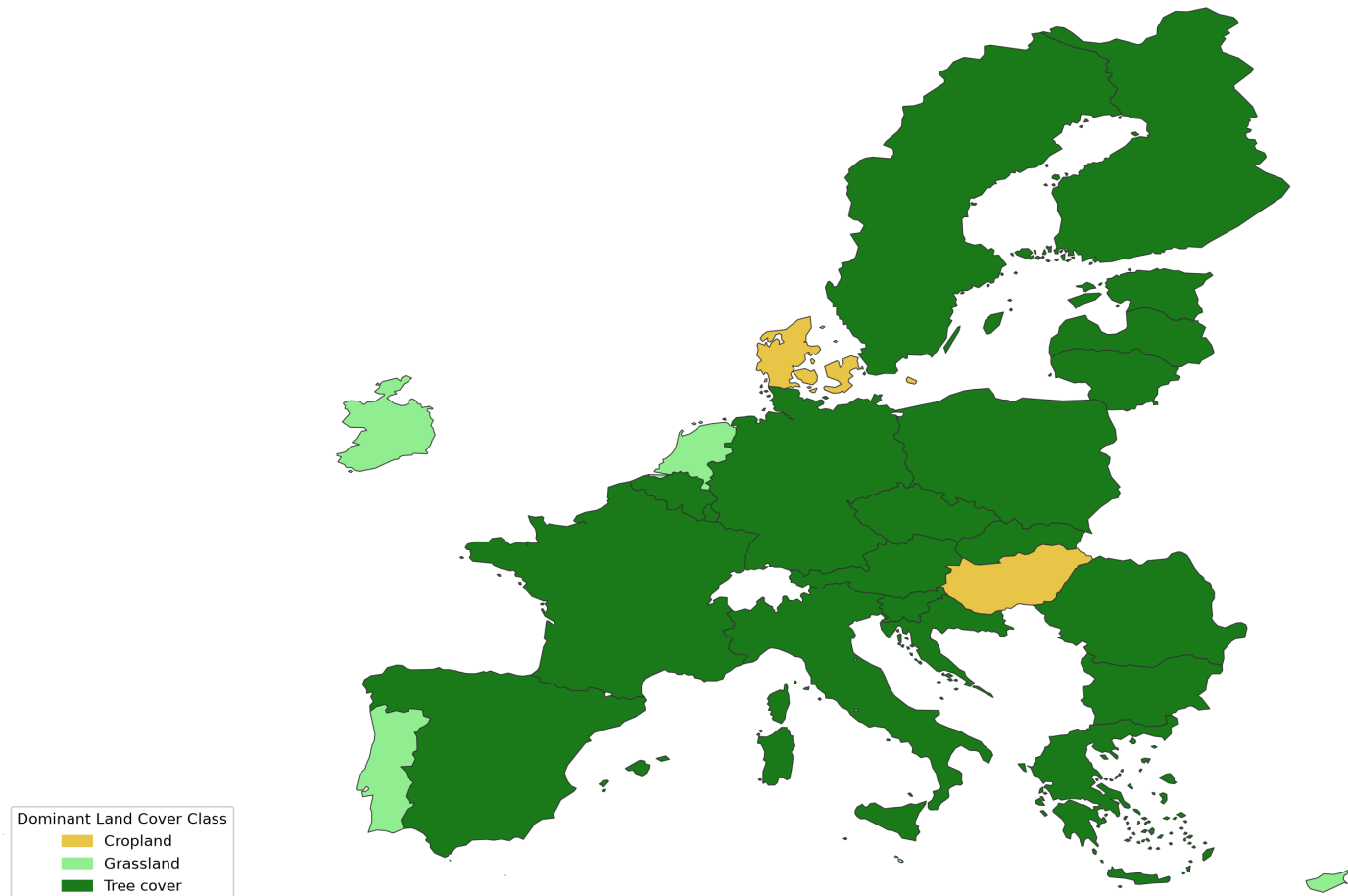
Green Policy Intelligence Engine (GPIE) — Source: Eurostat (EU-27), World Bank (control group)

GPIE - Green Policy Intelligence Engine

Dominant Land Cover Class Map

Control variable - dominant land-cover classification by country.

Dominant Land Cover Class by Country (EU-27)
Each country shaded by its single largest ESA WorldCover class (2021)



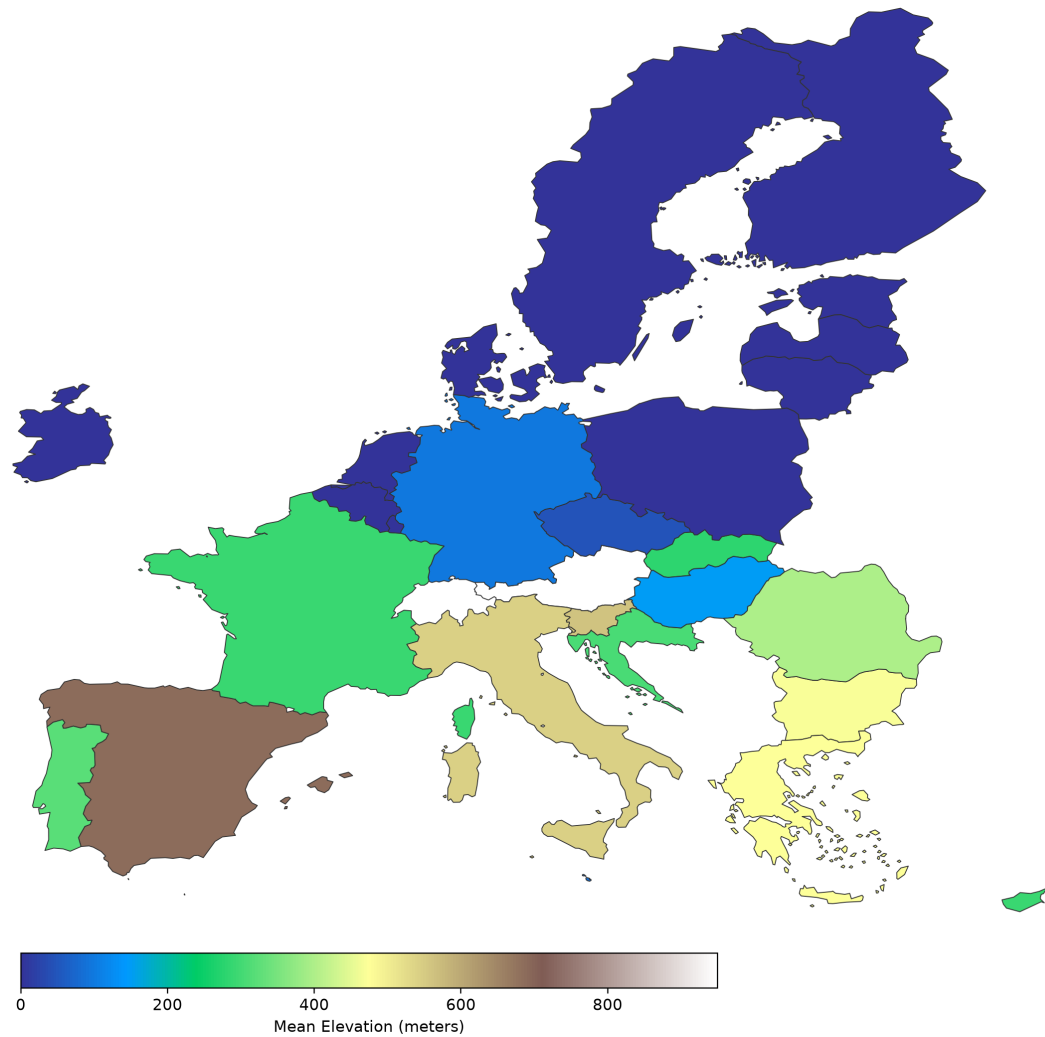
Green Policy Intelligence Engine (GPIE) — Source: ESA WorldCover 10m v200

GPIE - Green Policy Intelligence Engine

Elevation (DEM) Map

Control variable - digital elevation model.

Mean Elevation by Country (EU-27)
Environmental/topographic context — Copernicus DEM GLO-30

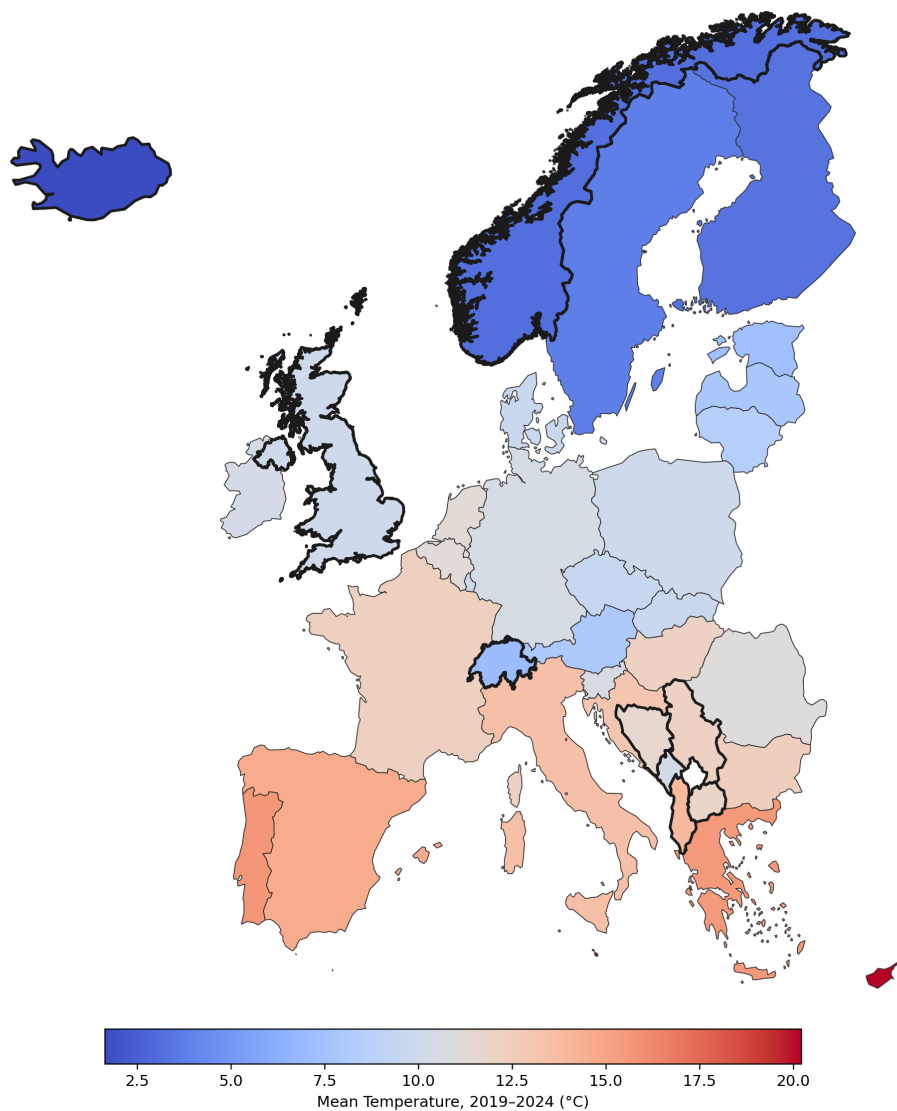


Climate / Temperature Map

Control variable - average temperature by country.

Mean Temperature: EU-27 vs. 9-Country Control Group

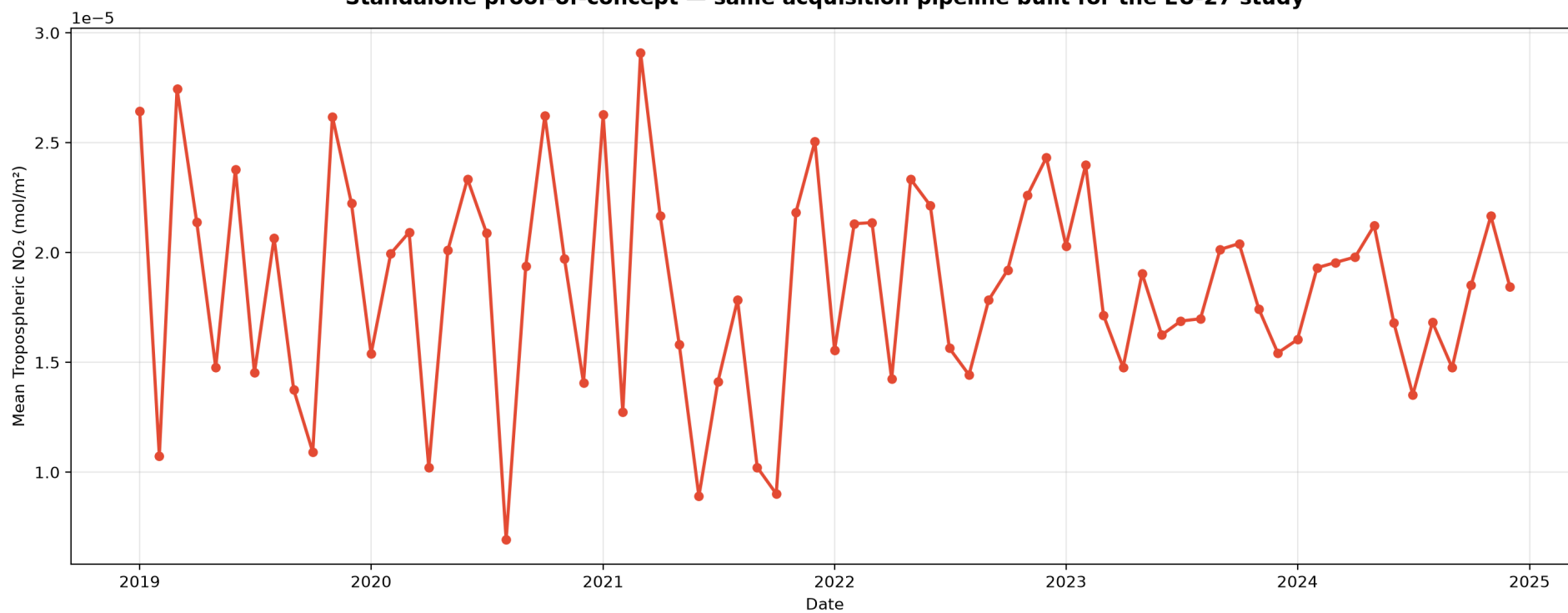
Thick borders mark non-EU control-group countries — temperature used as a climate control variable in the causal model



India Transferability Validation

NO2 acquisition pipeline independently tested on India, 2019-2024.

GPIE Transferability Test: India NO₂ Trend (2019-2024)
Standalone proof-of-concept — same acquisition pipeline built for the EU-27 study



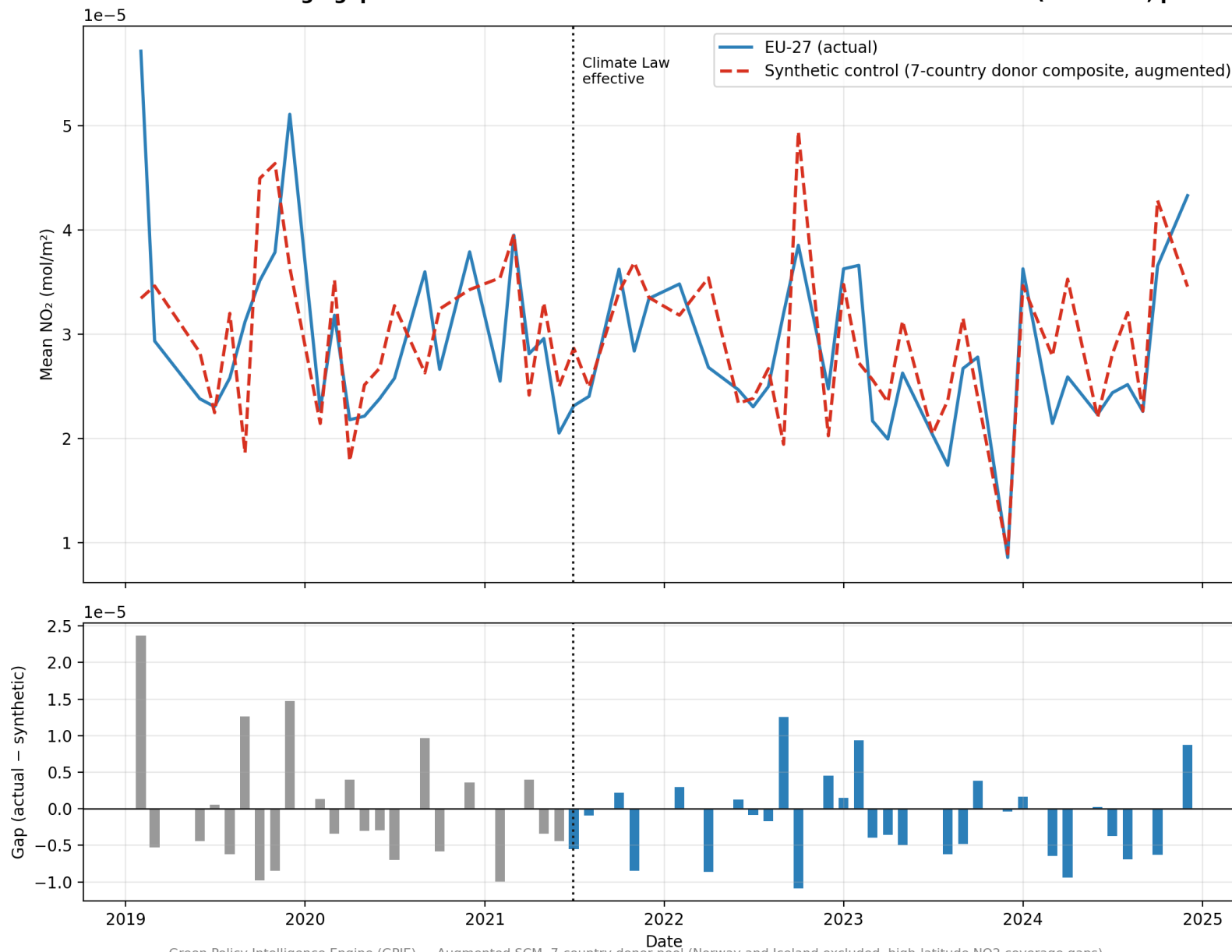
Green Policy Intelligence Engine (GPIE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

GPIE - Green Policy Intelligence Engine

Augmented Synthetic Control: EU-27 vs. 7-Country Donor Composite

eatment gap = $-1\text{e-}6$, same near-zero direction as the DiD estimate ($-2.22\text{e-}06$, $p=0.101$), reached independently via a 7-country donor pool (Norway and Iceland excluded, high-latitude NO₂ coverage

Augmented Synthetic Control: EU-27 vs. a 7-Country Weighted Composite Post-treatment average gap = -0.000001 — same near-zero direction as the DiD estimate ($-2.22\text{e-}06$, $p=0.101$)

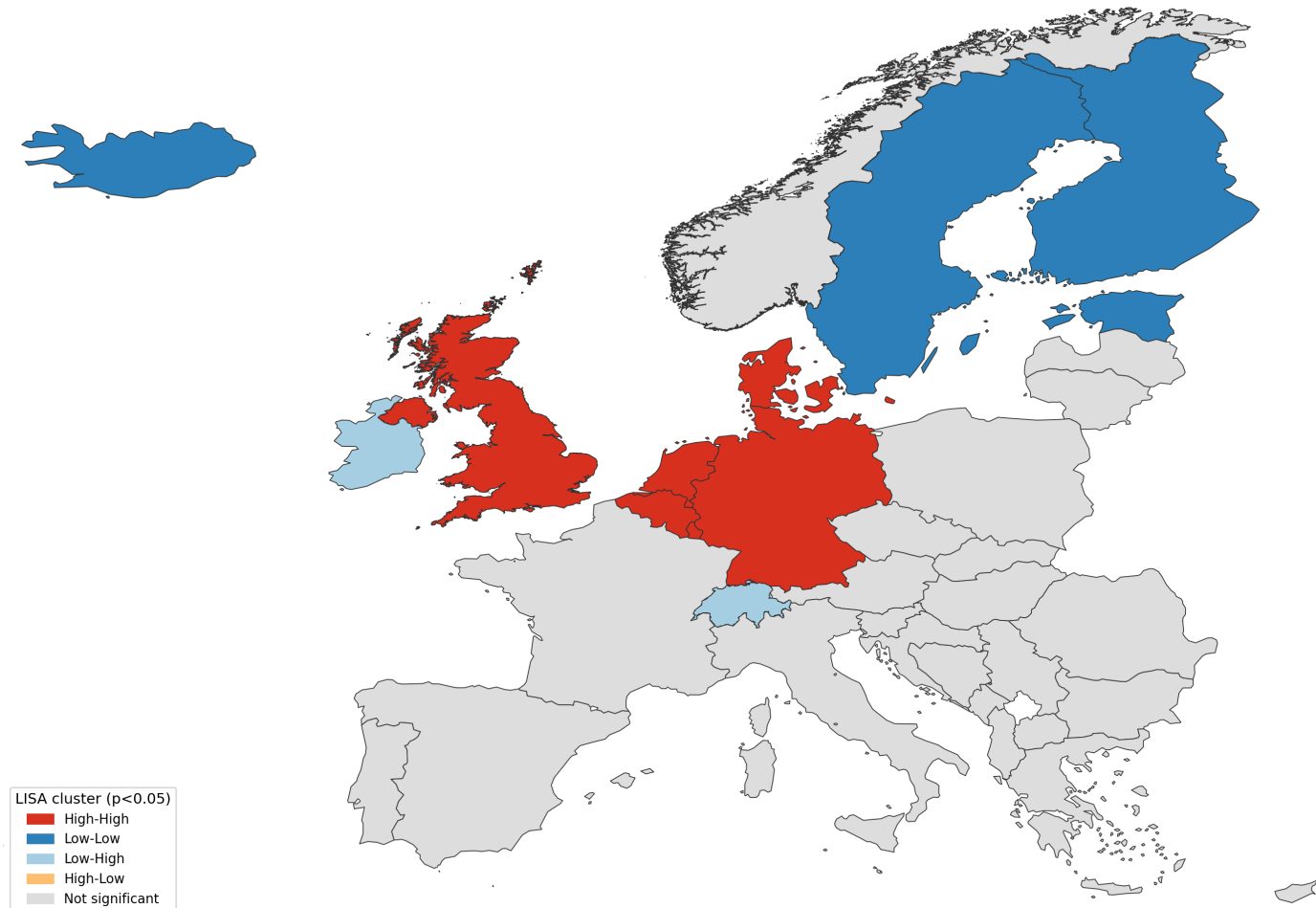


Green Policy Intelligence Engine (GPIE) — Augmented SCM, 7-country donor pool (Norway and Iceland excluded, high-latitude NO₂ coverage gaps)

Local Moran's I (LISA) Spatial Cluster Map

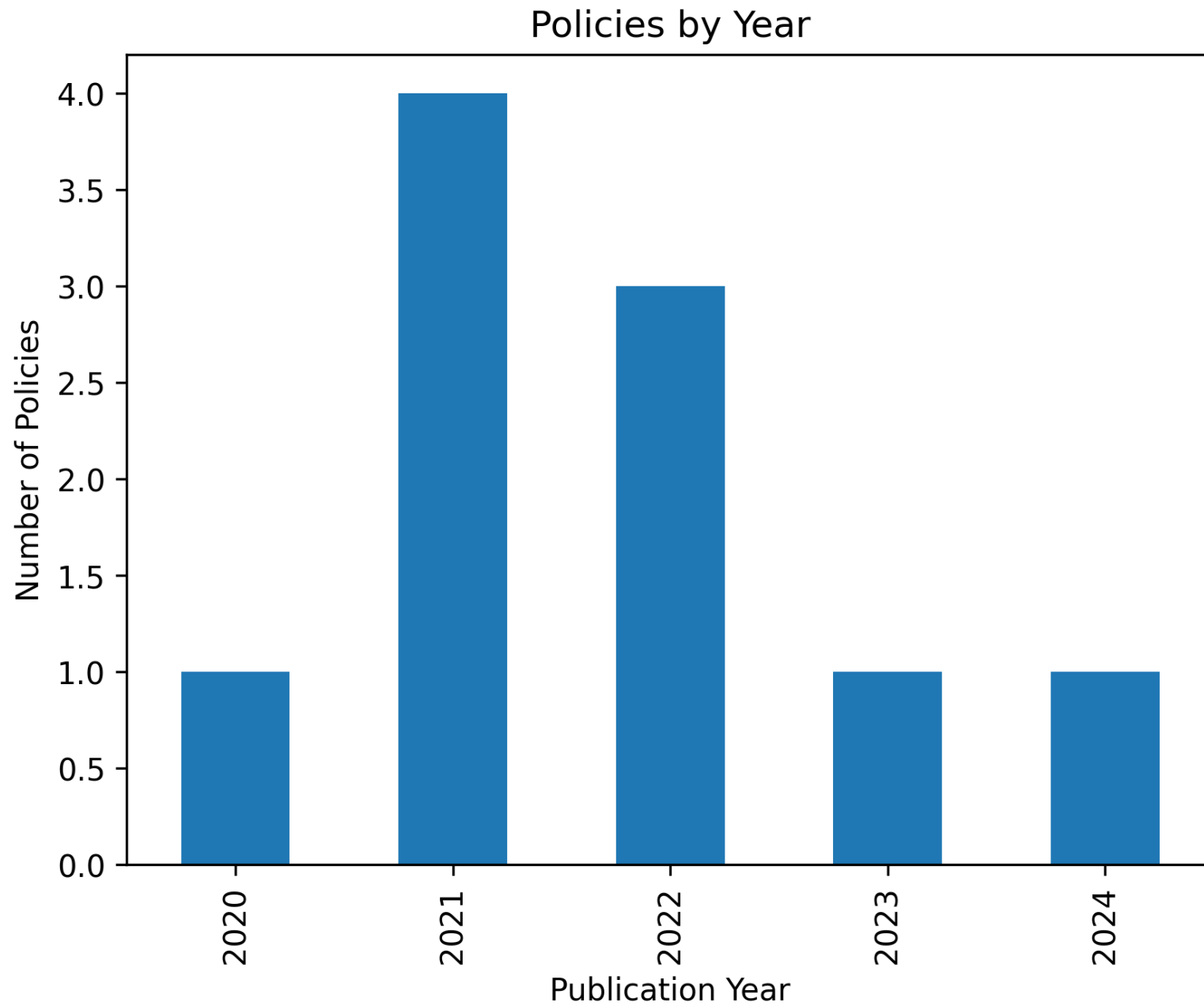
NO₂ levels cluster spatially across 36 countries ($I=0.570$, $p=0.001$); DiD residuals do not ($I=0.069$, $p=0.135$) - fixed effects absorb it.

Local Moran's I: Where NO₂ Levels Cluster Spatially (Full-Period Average, 36 Countries)
Global Moran's I = 0.570, $p = 0.001$ — pollution levels are strongly spatially clustered;
DiD model residuals are not ($I = 0.069$, $p = 0.135$), consistent with fixed effects absorbing it



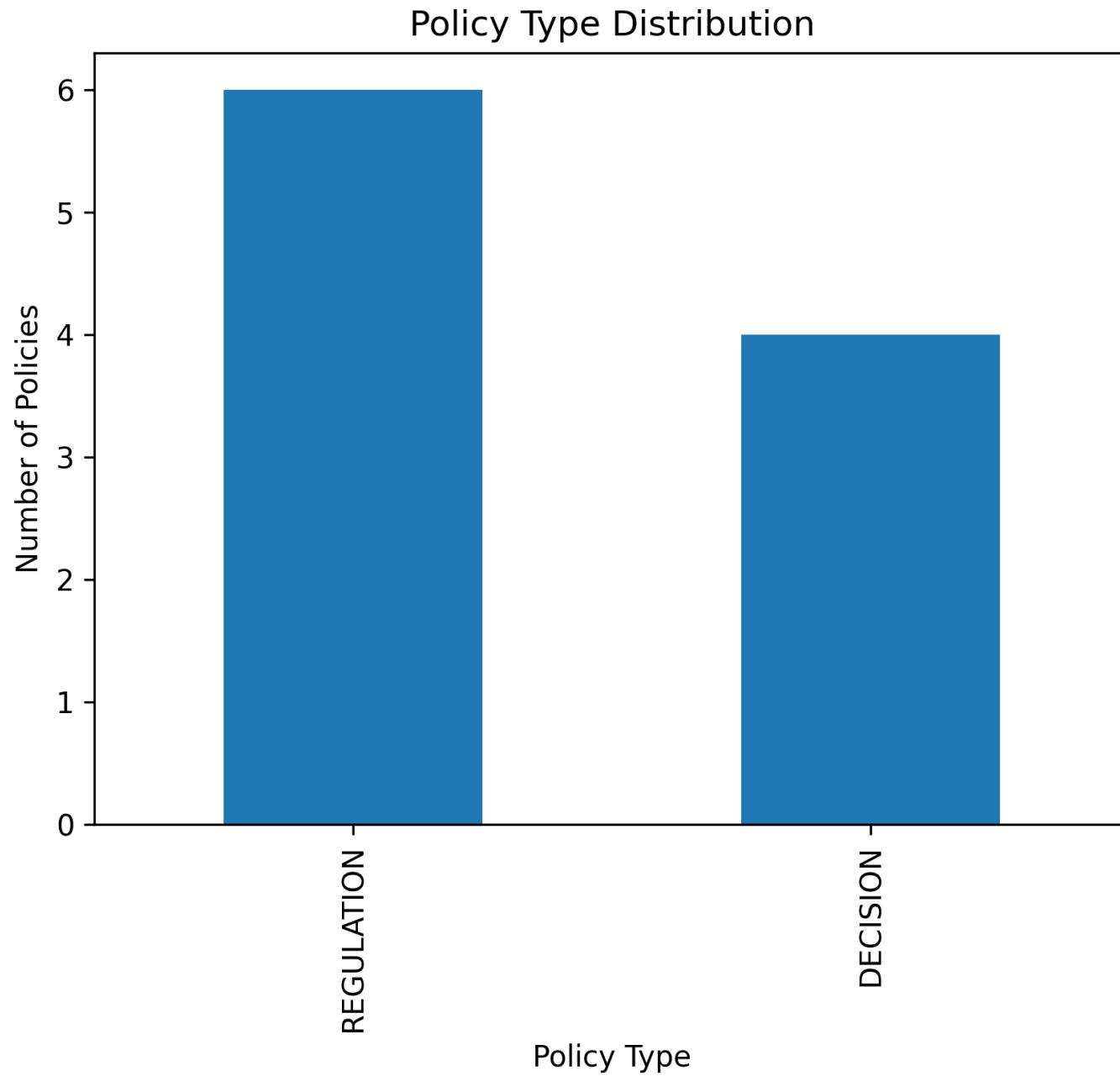
Policies by Year

Count of EU environmental/climate policies enacted per year.



Policy Type Distribution

Breakdown of policy dataset by policy type.



Policy Types by Year

Policy type composition over time.

